EE2001 Electrical Circuits- I:

Continuation of the Topic:

Response of First-Order RL and RC Circuits

25 December 2023

Outline



Continuation of the topic:

5. Sequential Switching

6. Unbounded Response

7. The Integrating Amplifier

Additional Examples

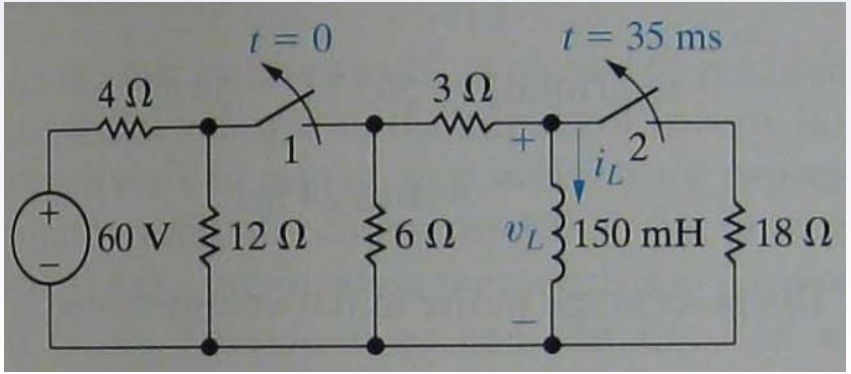
5 Sequential Switching

- Whenever switchin occures more than once in a circuit, we have sequential switching.
- A two-position switch may be switched back and forth, or multiple switches may be opened or closed in sequence.
- The time reference for all switchings cannot be t=0
- We use the techniques described previously, by deriving the expressions for $v(t)$ and $i(t)$ for a given position of the switch or switches and then use these solutions to determine the initial conditions for the next position of the switch or switches.
- It is important to obtain the initial value of a variable. Please recall that anything but inductive current and capacitive voltages can change instantaneously at the time of switching. Thus solving first for inductive current and capacitive voltages is even more suitable in sequential switching problems.
- Drawing the circuit for each time interval is often helpful in the solution process.

The Step and the Total Responses of RL and RC Circuits

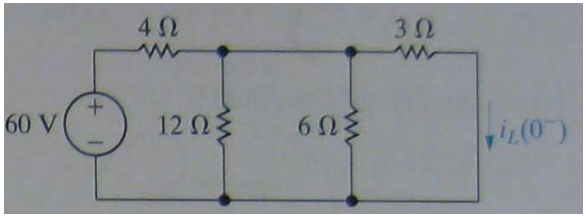
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- Example-8: The two switches in the given circuit have been close for a long time. At $t = 0$, switch 1 is opened. Then, 35 ms later, switch 2 is opened



a) Find $i_L(t)$ for $0 \leq t \leq 35\ ms$

Solution: a) For $t < 0$ both switches are closed, causing the 150 mH inductor to short circuit the 18 Ω resistor. The equivalent circuit is:



We determine the initial current in the inductor by solving for $i_L(0^-)$. After several sources transformations, we find it as 6 A.

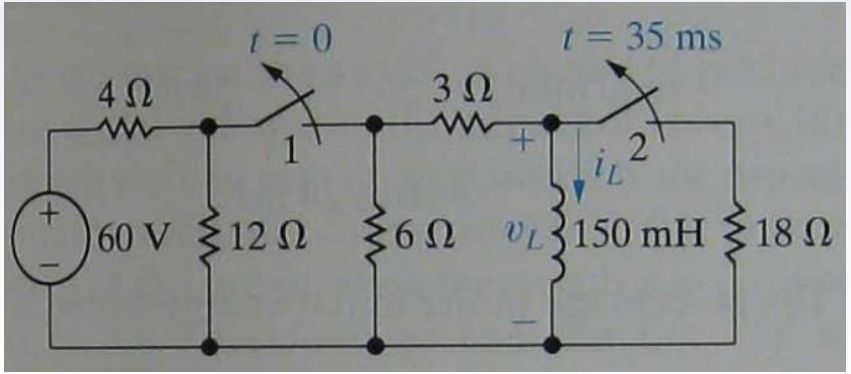
For $0 \leq t \leq 35\ ms$, switch 1 is open (switch

2 is closed), which disconnects the 60 V

voltage source and the 4 Ω and 12 Ω resistors from the circuit.

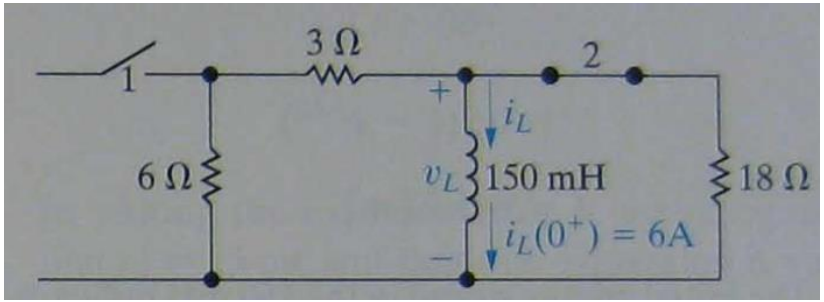
The Step and the Total Responses of RL and RC Circuits

- Example-8: The two switches in the given circuit have been close for a long time. At $t = 0$, switch 1 is opened. Then, 35 ms later, switch 2 is opened



a) Find $i_L(t)$ for $0 \leq t \leq 35\ ms$

Continuation of a) The inductor is no longer behaving as a short circuit (because the DC source is no longer in the circuit).



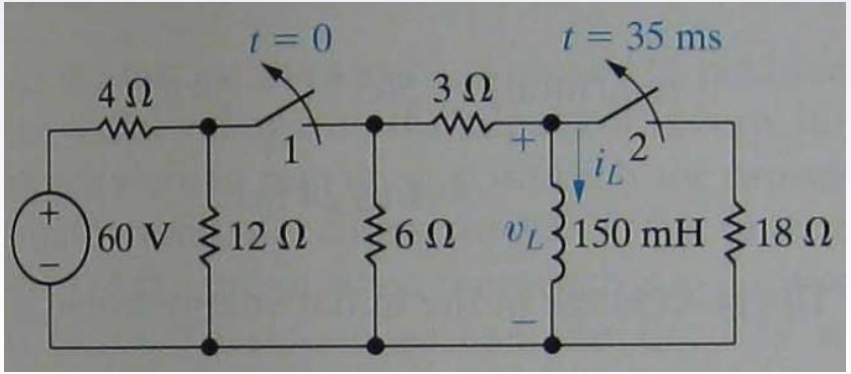
The 18 Ω resistor is no longer short-circuited.

Equivalent resistors across the terminals of the Inductor is the parallel combination of 9 Ω and 18 Ω; which is 6 Ω. The time constant of the circuit is $({}^{150}/_{6}) \times 10^{-3} = 25$ ms

Therefore, $i_L(t) = 6e^{-40t}\ A, \qquad 0 \leq t \leq 35\ ms$

The Step and the Total Responses of RL and RC Circuits

- Example-8: The two switches in the given circuit have been close for a long time. At $t = 0$, switch 1 is opened. Then, 35 ms later, switch 2 is opened

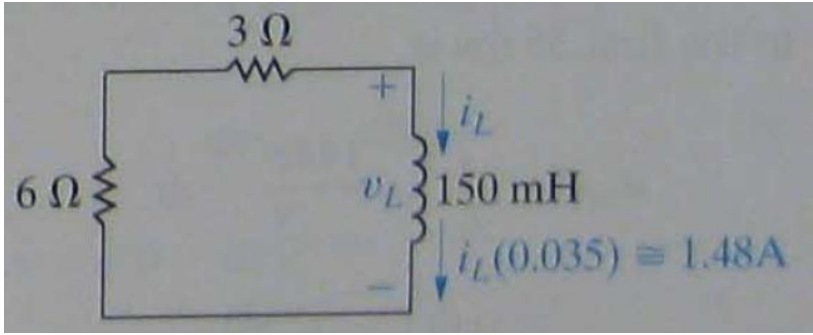


b) Find $i_L(t)$ for $t \geq 35\ ms$

b) When $t = 35\ ms$, the value of the inductor current is

$i_L = 6e^{-1.4} = 1.48\ A$

Thus, when switch 2 is opened, the circuit reduces to:



The time constant changes to

$$\tau = \left({}^{150}/_{9}\right) \times 10^{-3} = 16.67\ ms$$

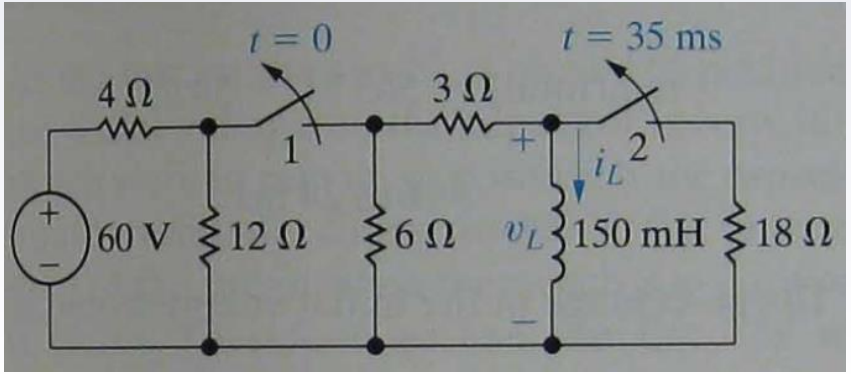
And $i_L(t) = 1.48e^{-60(t-0.035)}\ A, \qquad t \geq 35\ ms$

Note that the exponential function is shifted in time by 35 ms.

The Step and the Total Responses of RL and RC Circuits

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- Example-8: The two switches in the given circuit have been close for a long time. At $t = 0$, switch 1 is opened. Then, 35 ms later, switch 2 is opened



c) What percentage of the initial energy stored in the 150 mH inductor is dissipated in the 18 Ω resistor?

c) The 18 Ω resistor is in the circuit only during the first 35 ms of the switching sequence. During this interval, the voltage across the resistor is

$$v_L = 0.15\frac{d}{dt}(6e^{-40t}) = -36e^{-40t}\ V, \qquad 0 < t < 35\ ms$$

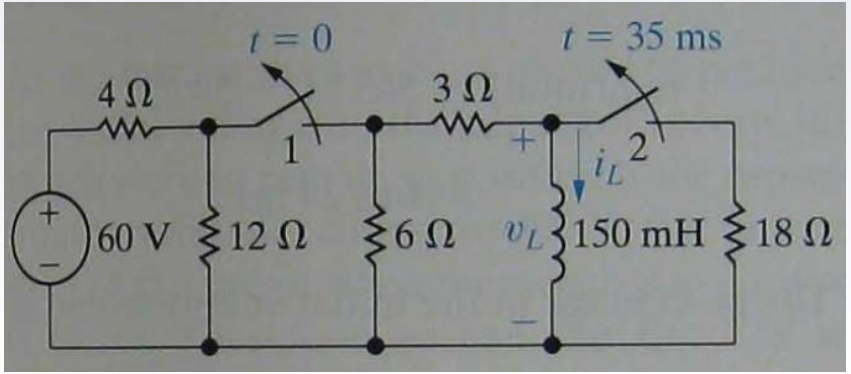
The power dissipated in the 18 Ω resistor is

$$p = \frac{{v_L}^2}{18} = 72e^{-80t}\ W, \qquad 0 < t < 35\ ms$$

The Step and the Total Responses of RL and RC Circuits

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- Example-8: The two switches in the given circuit have been close for a long time. At $t = 0$, switch 1 is opened. Then, 35 ms later, switch 2 is opened



c) What percentage of the initial energy stored in the 150 mH inductor is dissipated in the 18 Ω resistor?

Continuation of c) The energy dissipated is

$$w = \int_0^{0.035} 72e^{-80t}dt = \frac{72}{80}e^{-80t}\Big|_0^{0.035} = 0.9(1 - e^{-2.8}) = 845.27\ mJ$$

The initial energy stored in the 150 mH inductor is $w_i = \frac{1}{2}(0.15) \times 36) = 2.7\ J = 2700\ mJ$

$(^{845.27}/_{2700}) \times 100 = 31.31\%$ of the initial energy stored in the inductor is dissipated in the 18 Ω resistor.

Unbounded Response

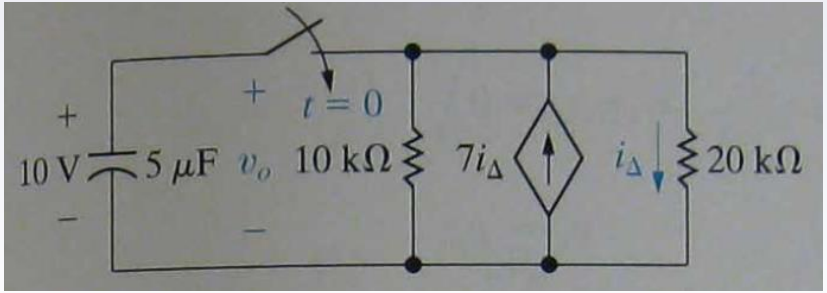
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- A circuit response may grow, rather than decay, exponentially with time.
- This type of response, called an unbounded response, is possible if the circuit contains dependent sources. In that case, the Thevenin resistance with respect to terminals of either an inductor or a capacitor may be negative.
- Negative resistance generates a negative time constant, and the resulting currents and voltages increase without limit.
- In an actual circuit, the response eventually reaches a limiting value when a component breaks downor goes into a saturation state, prohibiting further increases in voltage or current.
- In unbounded response cases, the concept of a final value is confusing. Hence, rather than using the step response solution, we derive the differential equation that describes the circuit containing the negative resistance and then solve it using the seperation of variables technique.

Unbounded Response

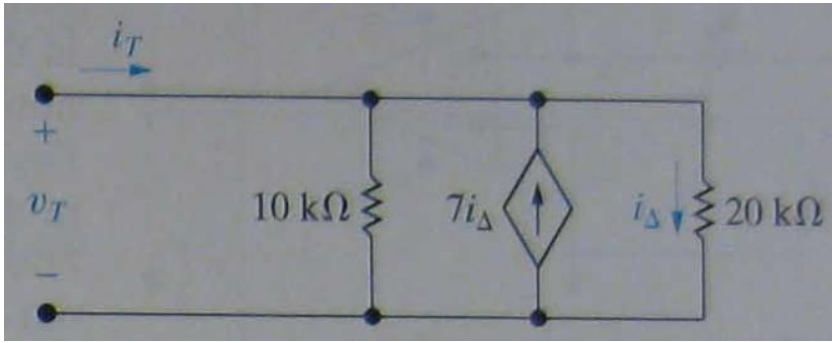
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Example-9: a) When the switch closed in the given circuit, the voltage on the capacitor is 10 V. Find the expression for $v_0(t)$ for $t \geq 0$



b) Assume that the capacitor short-circuits when its terminal voltage reaches 150 V. How many miliseconds elapse before the capacitor short-circuits.

Solution: a) To find the Thevenin equivalen resistance with respect to the capacitor terminals, we use the test-source method from previos topics.



v_T is the test voltage and i_T is the test current.

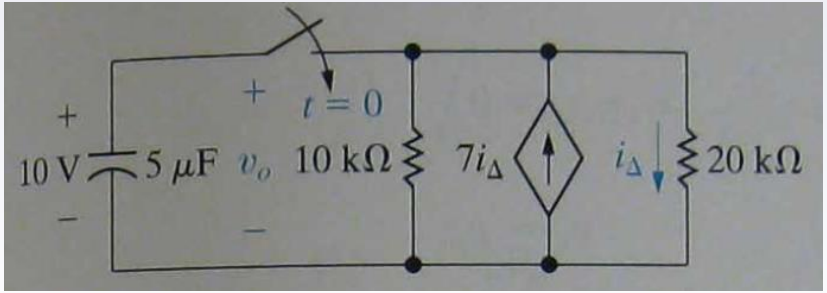
$$i_T = \frac{v_T}{10} - 7\left(\frac{v_T}{20}\right) + \frac{v_T}{20}\ mA$$

Solving for the ratio ${}^{v_T}/_{i_T}$ yields $R_{Th} = -5\ k\Omega$

Unbounded Response

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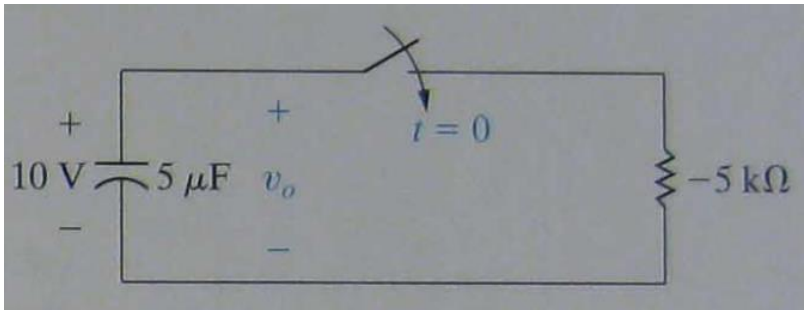
Example-9: a) When the switch closed in the given circuit, the voltage on the capacitor is 10 V. Find the expression for $v_0(t)$ for $t \geq 0$



b) Assume that the capacitor short-circuits when its terminal voltage reaches 150 V. How many miliseconds elapse before the capacitor short-circuits.

Continuation of a) For $t \geq 0$ the differential equation describing the simplified circuit

$$(5 \times 10^{-6})\frac{dv_0}{dt} - \frac{v_0}{5}10^{-3} = 0$$



Dividing by the coefficient of the first derivative yields:

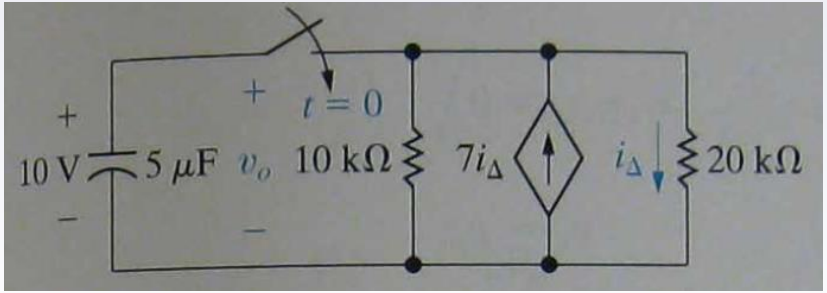
$$\frac{dv_0}{dt} - 40v_0 = 0$$

Using the seperation of variables technique: $v_0(t) = 10e^{40t}\ V, \quad t \geq 0$

Unbounded Response

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Example-9: a) When the switch closed in the given circuit, the voltage on the capacitor is 10 V. Find the expression for $v_0(t)$ for $t \geq 0$

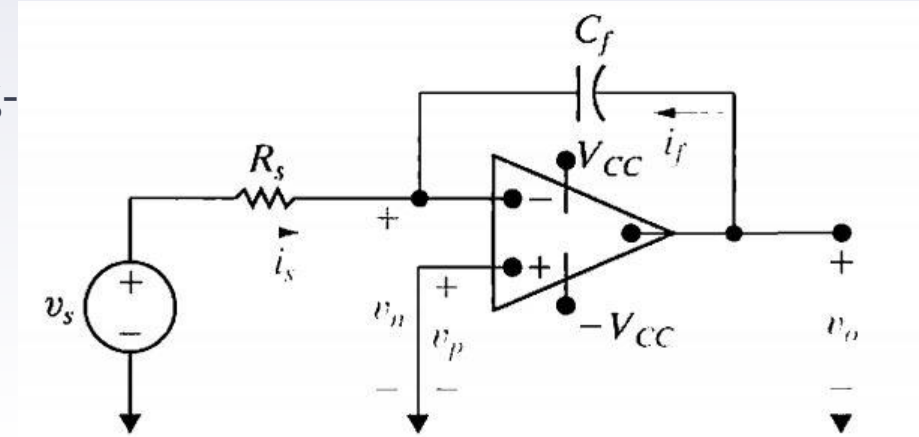


b) Assume that the capacitor short-circuits when its terminal voltage reaches 150 V. How many miliseconds elapse before the capacitor short-circuits.

b) $v_0 = 150\ V$ when $e^{40t} = 15$. Therefore $40t = ln15$ and $t = 67.7\ ms$

The Integrating Amplifier

- We are now ready to analyze an integrating-amplifier circuit.
- It generates an output voltage proportional to the integral of the input voltage.
- Taking into account the branch currents i_f and i_s .



- Assuming that the operational amplifier is ideal,

$$i_f + i_s = 0$$

$$v_n = v_p = 0$$

- We can write,

$$i_s = \frac{v_s}{R_s}$$

$$i_f = C_f \frac{dv_o}{dt}$$

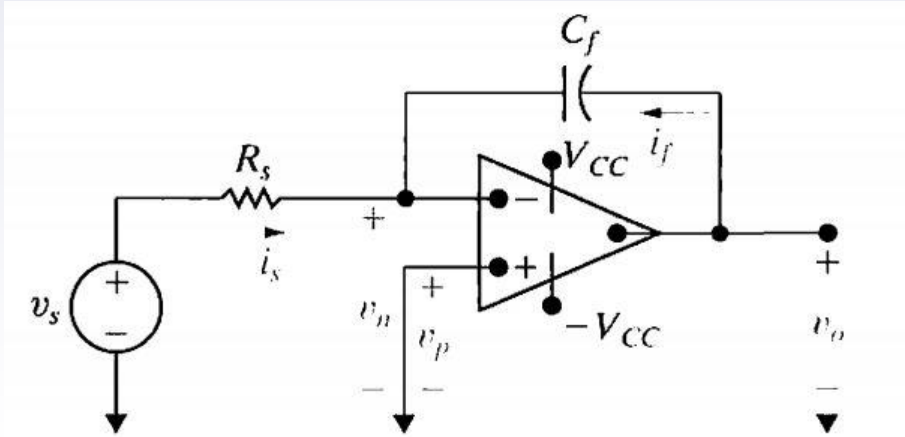
The Integrating Amplifier

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- Organizing the equations,

$$\frac{dv_o}{dt} = -\frac{1}{R_s C_f} v_s$$

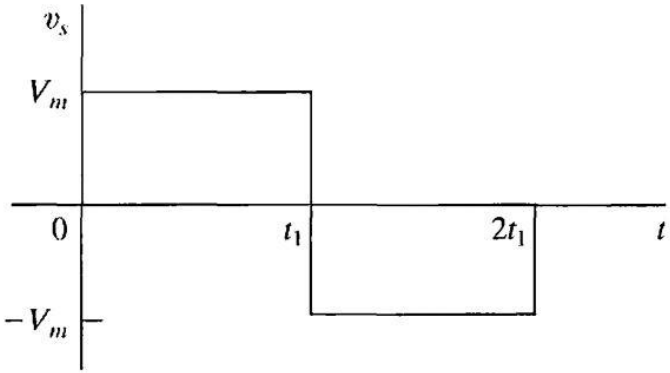
$$v_o(t) = -\frac{1}{R_s C_f} \int_{t_0}^{t} v_s dy + v_o(t_0)$$



- If no energy is stored in the capacitor when integration commences, the equation reduces to:

$$v_o(t) = -\frac{1}{R_s C_f} \int_{t_0}^{t} v_s dy$$

- If v_s is a step change in a DC voltage level, the output voltage will vary linearly with time. For example, assume that the input voltage is the rectangular voltage pulse



The Integrating Amplifier

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- Also assuming that the initial value of $v_o(t)$ is zero at the instant v_s steps from 0 to V_m:

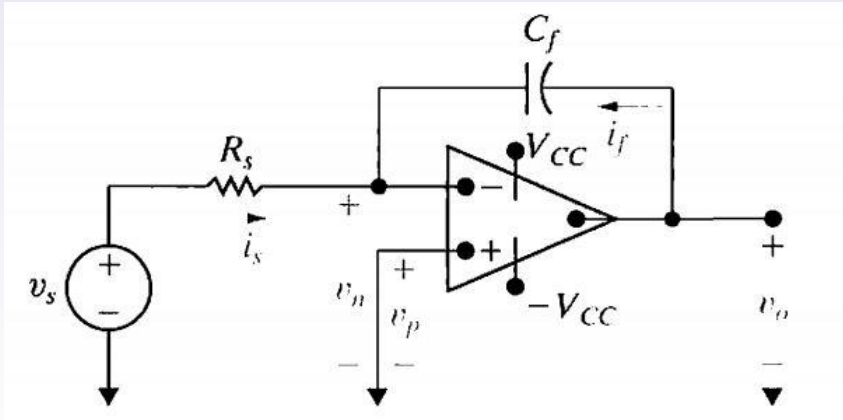
$$v_o(t) = -\frac{1}{R_s C_f} V_m t + 0, \qquad 0 \le t \le t_1$$

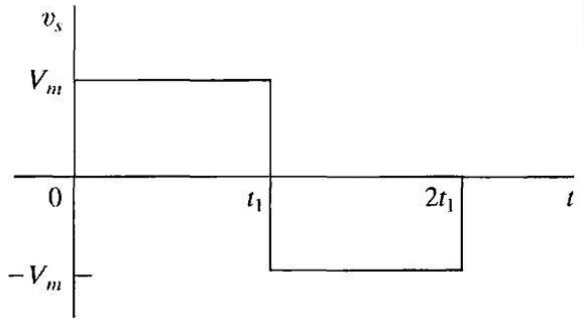
- When t lies between t_1 and $2t_1$,

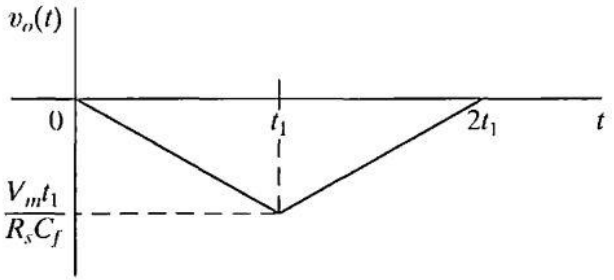
$$v_o = -\frac{1}{R_s C_f}\int_{t_1}^{t}(-V_m)dy - \frac{1}{R_s C_f} V_m t_1$$

$$= \frac{V_m}{R_s C_f} t - \frac{2V_m}{R_s C_f} t_1, \qquad t_1 \le t \le t_2$$

- The output voltage is proportional to the integral of the input voltage only if the op-amp operates within its linear range and does not saturate.

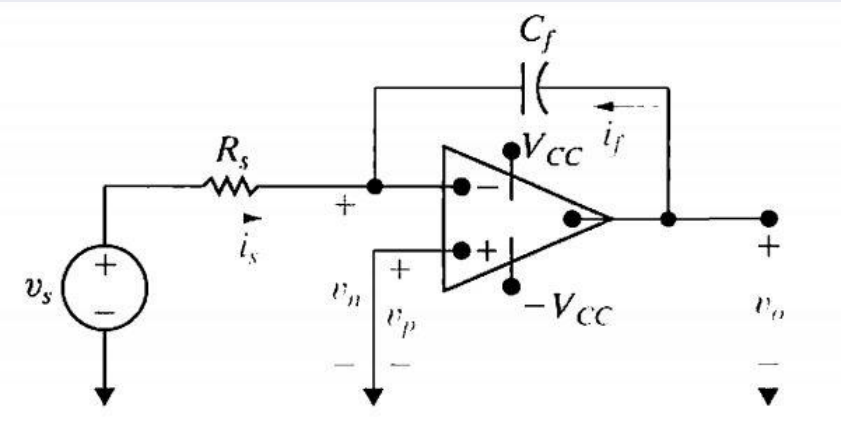


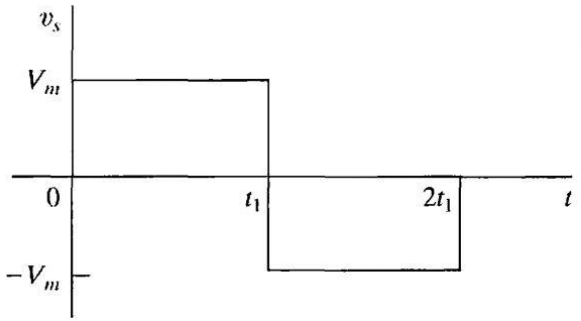




The Integrating Amplifier

- Example-10: Assume that the numerical values for the signal are $V_m = 50\ mV$ and $t_1 = 1\ s$. This signal voltage is applied to the integrating-amplifier circuit. The circuit parameters of the amplifier are $R_s = 100\ k\Omega$, $C_f = 0.1\ \mu F$ and V_{CC} =6 V. The initial voltage on the capacitor is zero.





a) Calculate $v_o(t)$.

b) Plot $v_o(t)$ versus t.

Solution: a) For $0 \le t \le 1\ s$,

$$v_o = \frac{-1}{(100 \times 10^3)(0.1 \times 10^{-6})} 50 \times 10^{-3} t + 0$$

$$= -5t\ V, \qquad 0 \le t \le 1\ s$$

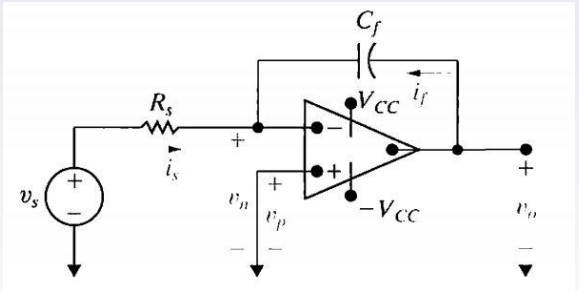
$$v_o = 5t - 10, \qquad 1 \le t \le 2\ s$$

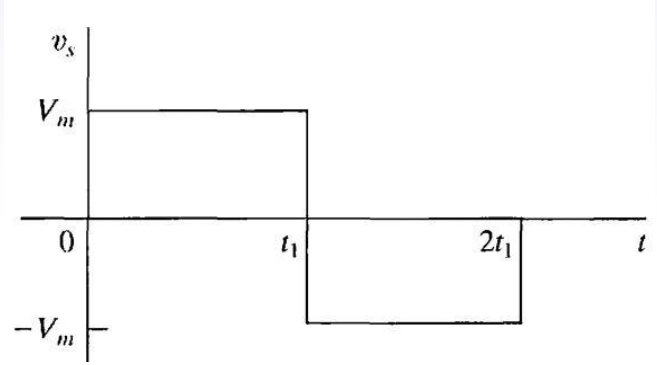
The Integrating Amplifier

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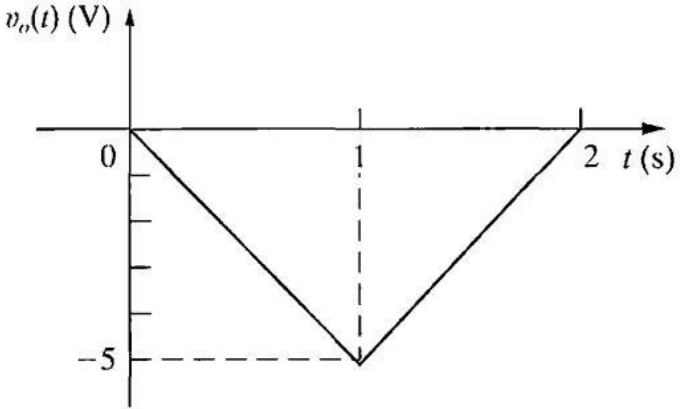
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b) Plot $v_o(t)$ versus t.





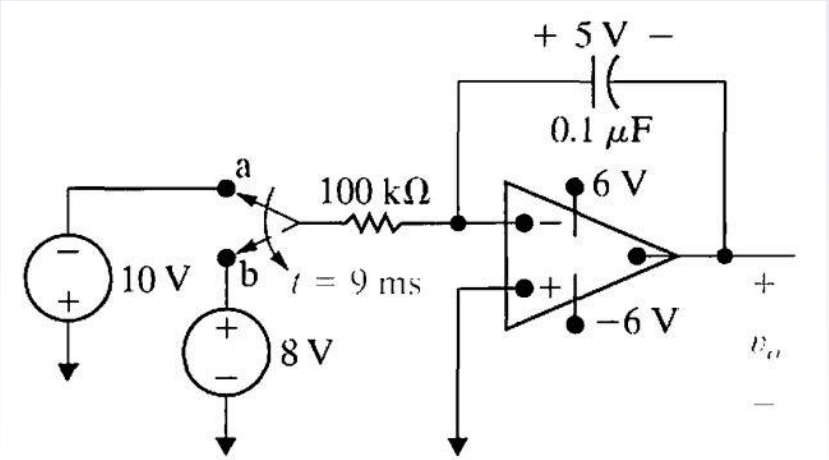
Solution: b)



The Integrating Amplifier

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- Example-11: At the instant the switch makes contact with terminal a in the given circuit, the voltage on the 0.1 μF capacitor is 5 V. The switch remains at terminal a for 9 ms and then moves instantaneously to terminal b. How many milisecondes after making contact with terminal b does the op-amp saturate?



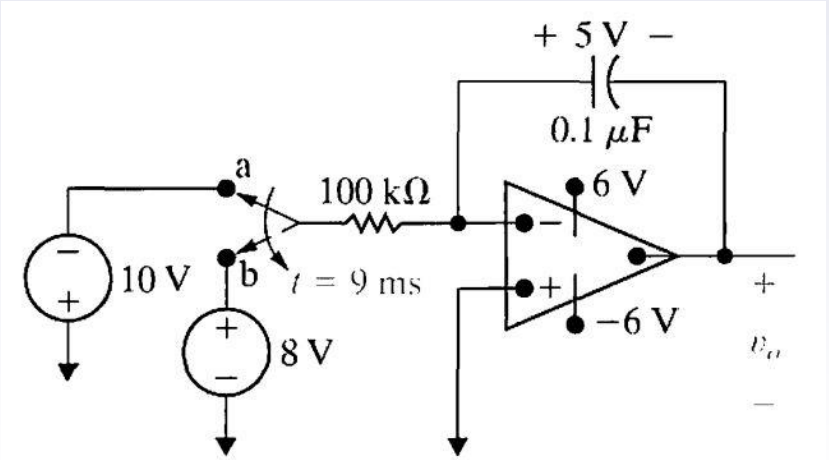
Solution: The expression for the output voltage during the time the switch is at terminal a is

$$v_o = -5 - \frac{1}{10^{-2}} \int_0^t (-10)dy$$
$$= (-5 + 1000t)\ V$$

9 ms after the switch makes contact with terminal a, the output voltage is $-5 + 9 = 4\ V$

The Integrating Amplifier

- Example-11: At the instant the switch makes contact with terminal a in the given circuit, the voltage on the 0.1 μF capacitor is 5 V. The switch remains at terminal a for 9 ms and then moves instantaneously to terminal b. How many milisecondes after making contact with terminal b does the op-amp saturate?



Solution: The expression for the output voltage after the time the switch moves to terminal b is:

$$v_o = 4 - \frac{1}{10^{-2}} \int_{9\times10^{-3}}^{t} 8dy$$

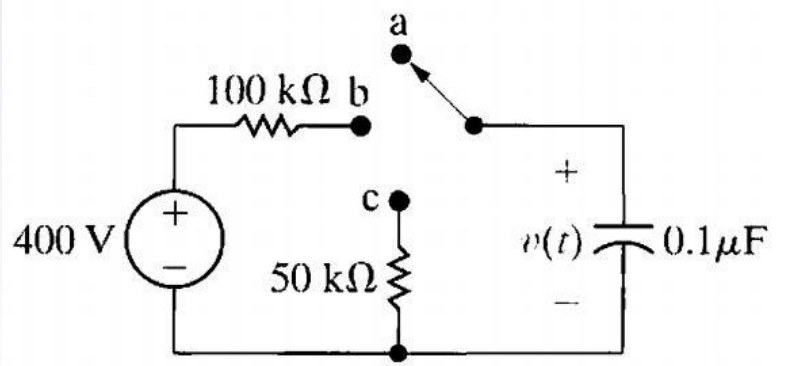
$$= 4 - 800(t - 9 \times 10^{-3}) = 11.2 - 800t\ V$$

During this time interval, the voltage is decreasing and the operational amplifier eventually saturates at $-6\ V$. By setting the expression for v_o equal to $-6\ V$, we can obtaint the saturation time: $11.2 - 800t_s = -6 \qquad t_s = 21.5\ ms$

Additional Examples

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- Example-12: The uncharged capacitor in the given circuit is initially switched to terminal a of the three-position switch. At $t = 0$, the switch is moved to position b, where it remains for 15 ms. After the 15 ms delay, the switch is moved to position c, where it remains indefinitely.



a) Derive the expression for the capacitor voltage.

Solution: a) At the instant the switch is moved to position b, the initial voltage on the capacitor is zero. If the swtich were to remain in position b, the capacitor would eventually charge to 400 V. The time constant of the circuit, when the switch is in position b is

$$\tau = RC = (100 \times 10^3)(0.1 \times 10^{-6}) = 10 \times 10^{-3}\, s = 10\, ms$$

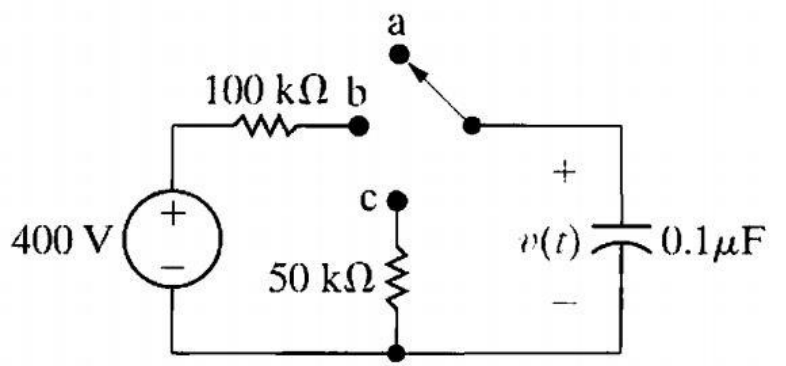
Using the general expression for the capacitor voltage:

$$v(t) = 400 + (0 - 400)e^{-100t} = (400 - 400e^{-100t})\, V, \qquad 0 \le t \le 15\, ms$$

Additional Examples

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- Example-12: The uncharged capacitor in the given circuit is initially switched to terminal a of the three-position switch. At $t = 0$, the switch is moved to position b, where it remains for 15 ms. After the 15 ms delay, the switch is moved to position c, where it remains indefinitely.



a) Derive the expression for the capacitor voltage.

Continuation of a)

$$v(t) = 400 + (0 - 400)e^{-100t} = (400 - 400e^{-100t})\ V, \qquad 0 \le t \le 15\ ms$$

Note that, the switch remains in position b for only 15 mins. After 15 ms, the voltage on the capacitor will be

$$v(15\ ms) = 400 - 400e^{-1.5} = 310.75\ V$$

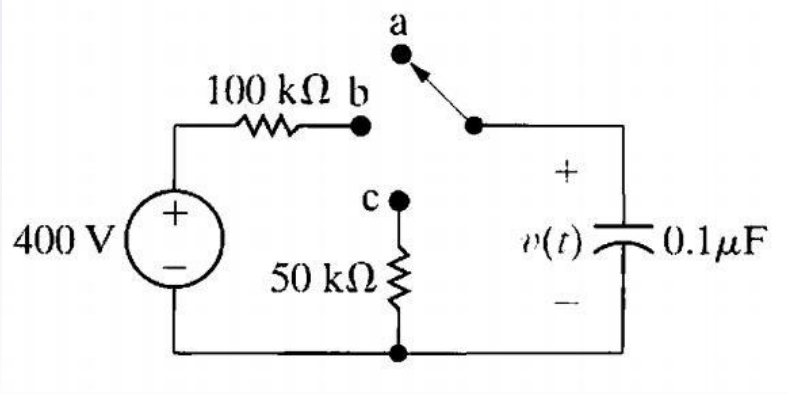
When the switch is moved to position c, this is the initial voltage on the capacitor 310.75 V.

The final value of the capacitor voltage is zero and the time constant is 5 ms.

Additional Examples

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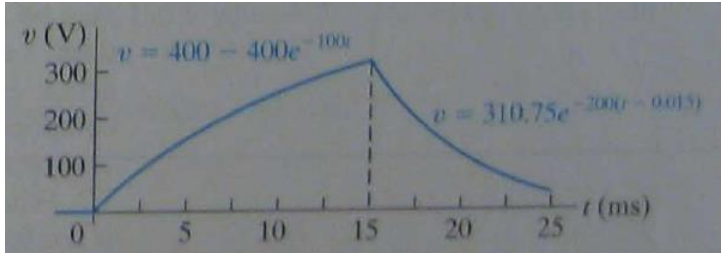
a) Derive the expression for the capacitor voltage.

b) Plot $v(t)$ versus t

Continuation of a) When the switch is in position c, the final value of the capacitor voltage is zero and the time constant is 5 ms.

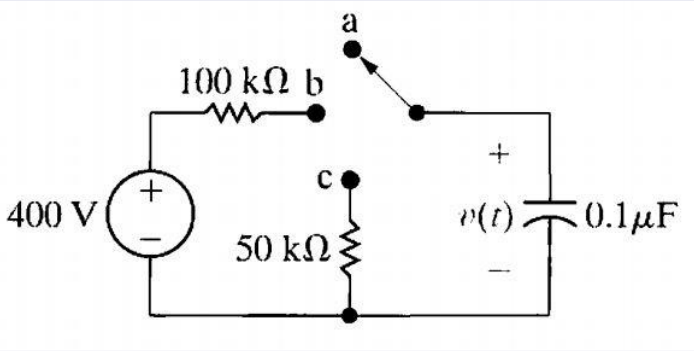
$$v(t) = 0 + (310.75 - 0)e^{-200(t-0.015)} = 310.75e^{-200(t-0.015)}\ V, \qquad 15\ ms \leq t$$

b)



Additional Examples

- Example-12: The uncharged capacitor in the given circuit is initially switched to terminal a of the three-position switch. At $t = 0$, the switch is moved to position b, where it remains for 15 ms. After the 15 ms delay, the switch is moved to position c, where it remains indefinitely.



c) When will the voltage on the capacitor equal 200 V?

c) The plot in the previous answer reveals that the capacitor wil equal 200 V at two differen times:

once in the interval between 0 and 15 ms and once after 15 ms.

We find the first time by solving the expression:

$$200 = 400 - 400e^{-100t_1} \qquad t_1 = 6.93\ ms$$

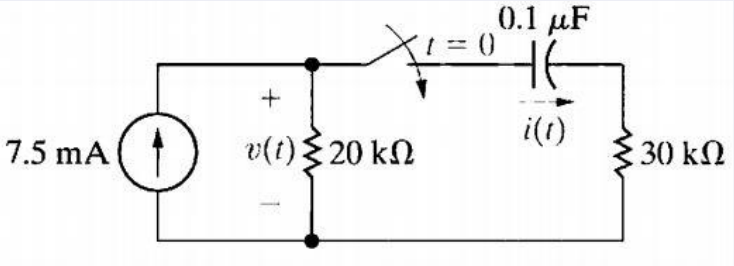
We find the second time by solving the expression:

$$200 = 310.75e^{-200(t_2 - 0.015)} \qquad t_2 = 17.20\ ms$$

Additional Examples

7

- Example-13: The switch in the given circuit has been open for a long time. The initial charge on the capacitor is zero. At $t = 0$, the switch is closed. Find the expression for



a) $i(t)$ for $t \geq 0^+$

a) Because the initial voltage on the capacitor is zero, at the instant when the switch is closed, the current in the 30 kΩ brach will be

$$i(0^+) = \frac{(7.5)(20)}{50} = 3\ mA$$

The final value of the capacitor current will be zero because the capacitor eventually will appear as an open circuit in terms of DC current. The time constant of the circuit will equal the product of the Thevenin resistance and the capacitance.

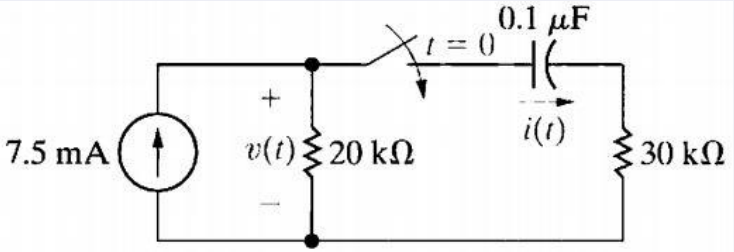
$$\tau = R_{th}C = (20 + 30)10^3(0.1) \times 10^{-6} = 5\ ms$$

$$i(t) = 0 + (3 - 0)e^{-t/5\times10^{-3}} = 3e^{-200t}\ mA, \qquad t \geq 0^+$$

Additional Examples

7

- Example-13: The switch in the given circuit has been open for a long time. The initial charge on the capacitor is zero. At $t = 0$, the switch is closed. Find the expression for



a) $i(t)$ for $t \geq 0^+$

b) $v(t)$ for $t \geq 0^+$

b) To find $v(t)$, we note from the circuit that it equals the sum of the voltage across the capacitor and the voltage across the 30 kΩ resistor. We note that the initial value is zero and the final value is $(7.5)(20) = 150\ V$. In the previous answer, we found the time constant as 5 ms. Using the general methodology the voltage expression for the capacitor:

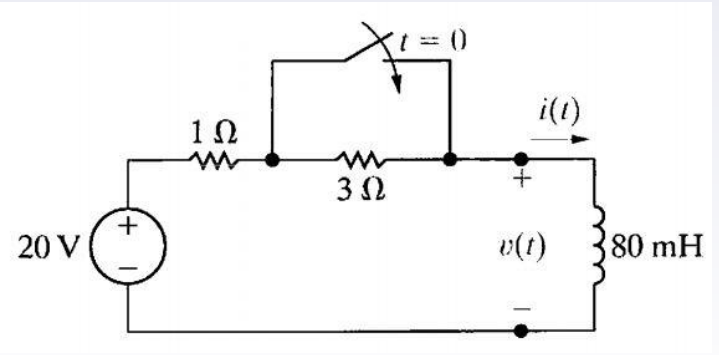
$$v_c(t) = 150 + (0 - 150)e^{-200t} = 150 - 150e^{-200t}\ V, \qquad t \geq 0$$

The expression for the voltage $v(t)$ is:

$$v(t) = 150 - 150e^{-200t} + (30)(3)e^{-200t} = 150 - 60e^{-200t}\ V, \quad t \geq 0^+$$

Additional Examples

- Example-14: The switch in the given circuit has been open for a long time. At $t = 0$, the switch is closed. Find the expression for



a) $v(t)$ for $t \geq 0^+$

b) $i(t)$ for $t \geq 0$

a) To switch has been open for a long time, so the initial current in the conductor is

$$20/(3+1) = 5\ A$$

20/4=5 A oriented from top to bottom. Immediately after the switch closes, the current still is 5 A, and therefore the initial voltage across the inductor becomes

$$20 - 5(1) = 15\ V$$

The final value of the inductor voltage is 0 V. The time constant is

$$L/R = (80/1) \times 10^{-3} = 80\ \text{ms}$$

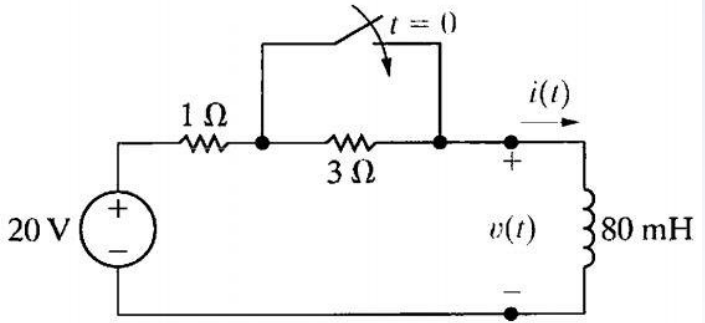
Using the general solution method:

$$v(t) = 0 + (15 - 0)e^{-\frac{t}{(80\times10^{-3})}} = 15e^{-12.5t}, \qquad t \geq 0^+$$

Additional Examples

7

- Example-14: The switch in the given circuit has been open for a long time. At $t = 0$, the switch is closed. Find the expression for



a) $v(t)$ for $t \geq 0^+$

b) $i(t)$ for $t \geq 0$

b) We have already noted that the initial value of the inductor current is 5 A. After the switch has been closed for a long time, the inductor current reaches $20/1 = 20\ A$. We also found that the time constant is 80 ms. The current expression $i(t)$ is

$$i(t) = 20 + (5 - 20)e^{-12.5t} = 20 - 15e^{-12.5t}\ A, \qquad t \geq 0$$

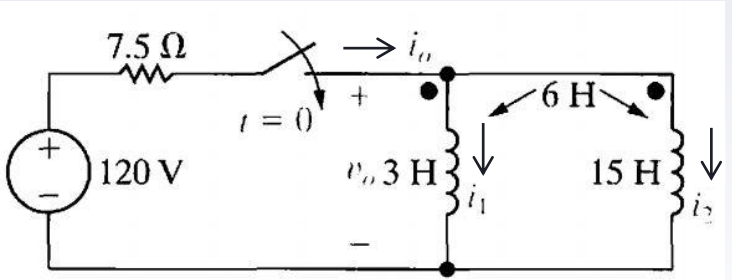
We check if $v(t)$ and $i(t)$ agree by noting that:

$$v(t) = L\frac{di(t)}{dt}$$
$$= 80 \times 10^{-3}[15(12.5)e^{-12.5t}]$$
$$= 15e^{-12.5t}\ V, \qquad t \geq 0^+$$

Additional Examples

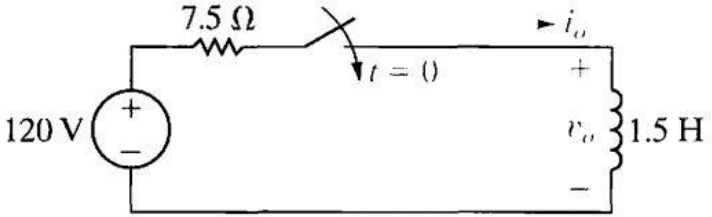
- Example-15: There is no energy stored in the given circuit at the time the switch is closed.

Find the solutions for i_o, v_o, i_1 and i_2.



For the circuit, the magnetically coupled coils can be replaced by a single inductor having an inductance of:

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} = \frac{45 - 36}{18 - 12} = 1.5\ H$$



The initial value of i_o is zero. The final value will be $120/7.5 = 16\ A$. The time constant is $1.5/7.5 = 0.2\ s$

$$i_o(t) = 16 + (0 - 16)e^{-5t} = 16 - 16e^{-5t}\ A, \qquad t \geq 0$$

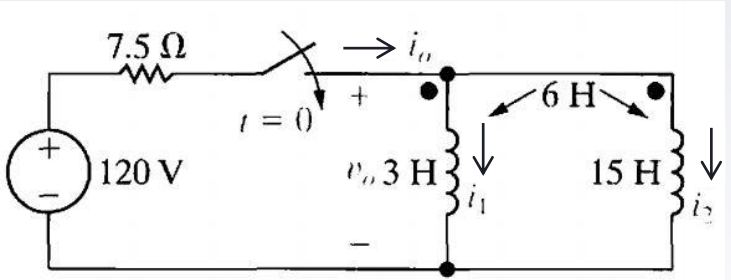
v_o follows from KVL:

$$v_o(\mathrm{t}) = 120 - 7.5 i_o(t) = 120e^{-5t}\ V, \qquad t \geq 0^+$$

Additional Examples

- Example-15: There is no energy stored in the given circuit at the time the switch is closed.

Find the solutions for i_o, v_o, i_1 and i_2.



Continuation of the solution: To find i_1 and i_2 we first note that

$$3\frac{di_1(t)}{dt} + 6\frac{di_2(t)}{dt} = 6\frac{di_1(t)}{dt} + 15\frac{di_2(t)}{dt} \qquad \frac{di_1(t)}{dt} = -3\frac{di_2(t)}{dt}$$

Also based on the relation $i_o = i_1 + i_2$

$$\frac{di_o(t)}{dt} = \frac{di_1(t)}{dt} + \frac{di_2(t)}{dt}$$

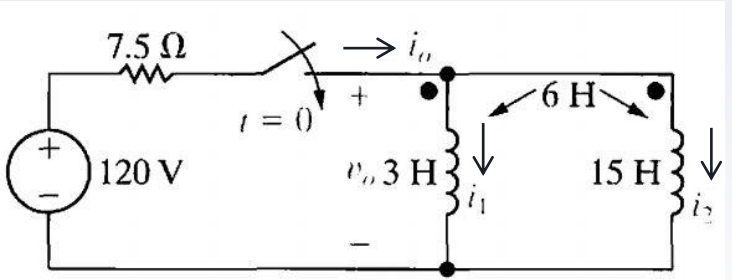
Therefore

$$80e^{-5t} = -2\frac{di_2(t)}{dt}$$

Additional Examples

- Example-15: There is no energy stored in the given circuit at the time the switch is closed.

Find the solutions for i_o, v_o, i_1 and i_2.



Continuation of the solution: Because $i_2(0)$ is zero we have

$$i_2 = \int_0^t -40e^{-5x}dx = -8 + 8e^{-5t}\ A, \qquad t \geq 0$$

Using KCL, we get

$$i_1 = 24 - 24e^{-5t}\ A, \qquad t \geq 0$$

EE2001 Electrical Circuits- I:

Sinusoidal Steady-State Analysis

25 January 2023

Outline



1. **The Sinusoidal Source**
2. **The Sinusoidal Response**
3. **The Phasor**
4. **The Passive Circuit Elements in the Frequency Domain**

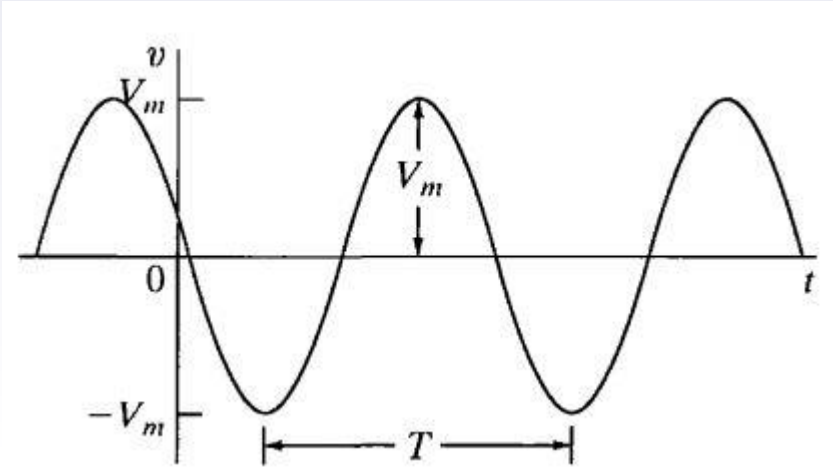
Introduction

- Since the beginning of the course, we have focused on circuits with constant sources.
- Now we will consider circuits energized by time-varying voltage or current sources.
- In particular, we are interested in sources in which the value of the voltage or current varies sinusoidally.
- Sinusoidal sources and their effect on circuit behavior form an important area of study.
- The generation, transmission, distribution and consumption of electric energy occur under essentially sinusoidal steady-state conditions.
    - Understanding of sinusoidal behavior makes it possible to predict the behavior of circuits with nonsinusoidal sources.
    - Steady-state sinusoidal behavior often simplifies the design of electrical systems.
    - A device, designed to satisfty steady-state sinusoidal specifications will respond satisfactorily to nonsinusoidal inputs.
- The challenges are to develop the equations and calculations with complex numbers.

The Sinusoidal Source

1

- A sinusoidal voltage source (independent or dependent) produces a voltage that varies sinusoidally with time.
- A sinusoidal currrent source (independent or dependent) produces a current that varies sinusoidally with time.



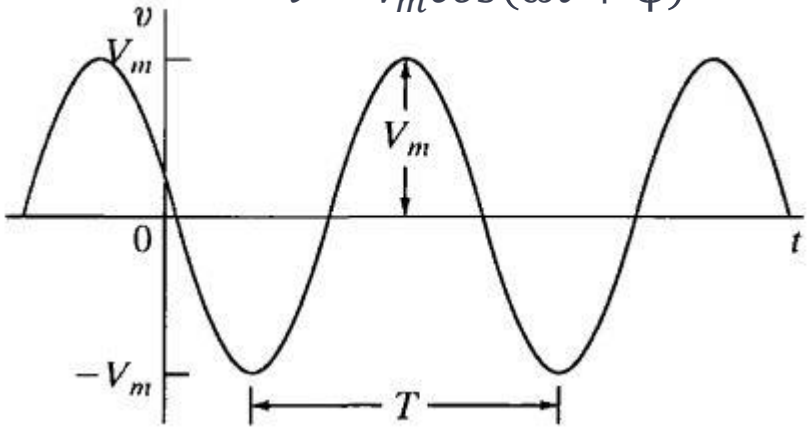
- We will see the formulations for a voltage, but the same applies to current sources.
- A sinusoidally varying function can be expressed with either the sine function or the cosine function. We will use the cosine function throughout our dicussion.
- Expression of a sinusoidally varying voltage:

$$v = V_m cos(\omega t + \phi)$$

- The function repeats at regular intervals. It is a periodic function.
- An important parameter is the time required for the function to pass through all its values.

The Sinusoidal Source

$v = V_m cos(\omega t + \phi)$



- This time is referred to as the period of the function and is denoted as T. It is measured in seconds.
- The reciprocal of T gives the number of cycles per second, or the frequency, of the sine functions and is denoted f.

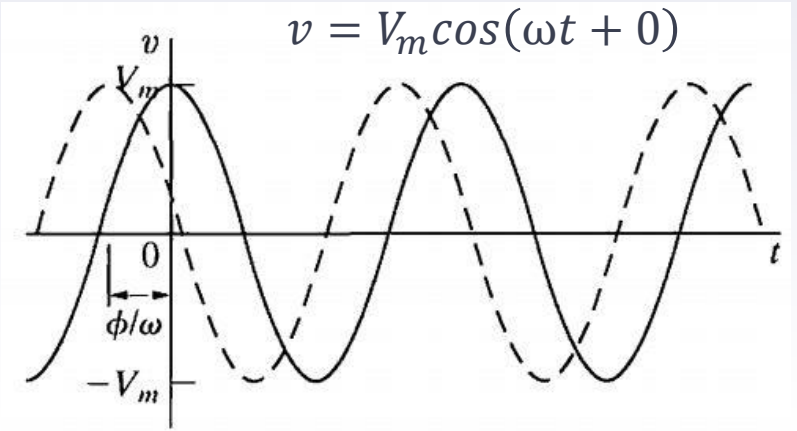
$$f = \frac{1}{T}$$

- The number of cycles per second is referred to as hertz, abbreviated as Hz.
- The coefficient of t in the sinusoidal function contains the numerical value of T or f. Omega (ω) represents the angular frequency of the sinusoidal function:

$$\omega = 2\pi f = 2\pi/T \quad \text{(radians/seconds)}$$

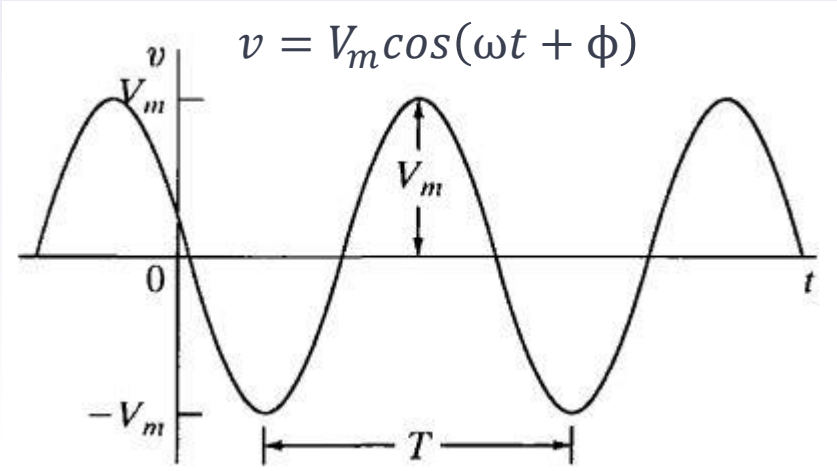
- ωt passes through 2π rad (360°).
- Whenever t is an integral multiple of T, ωt increases by an integral multiple of 2π rad.

The Sinusoidal Source



- The coefficient V_m gives the maximum amplitude of the sinusoidal voltage.
- Because ± 1 bounds the cosine function, $\pm V_m$ bounds the amplitude.
- The angle φ is known as the phase angle of the sinusoidal voltage.
- It determines the value of the sinusoidal function at $t = 0$. Therefore, it fixes the point on the periodic wave at which we start measuring time.
- Changing the phase angle φ shifts the sinusoidal function along the time axis, but has no effect on either the amplitude (V_m) or the angular frequency (ω).
- For example, reducing φ to zero shifts the sinusoidal function as shown in the figure (at time zero the signal reaches the positive peak value (V_m)).
- If φ is positive the sinusoidal function shifts to the left, if negative it shifts to the right.

The Sinusoidal Source



- ωt and φ must carry the same units, because they are added together.
- With ωt expressed in radians, you would expect φ to be also. However, φ normally is given in degrees and ωt is converted from radians to degrees before the two quantities are added.
- Conversion from radians to degrees:

$$(number\ of\ degrees) = \frac{180°}{\pi}\ (number\ of\ radians)$$

- Another important characteristic of the sinusoidal voltage (or current) is its rms value.
- It is the square root mean value of the squared function. For $v = V_m cos(\omega t + \varphi)$

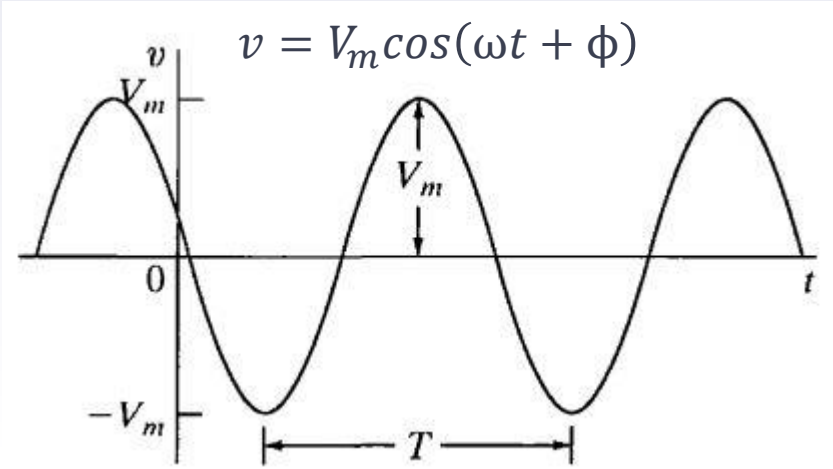
$$V_{rms} = \sqrt{\frac{1}{T}\int_{t_0}^{t_0+T} {V_m}^2 cos^2(\omega t + \varphi)dt}$$

The Sinusoidal Source

- The resulting equation is:

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

- The rms value of the sinusoidal signal depends only on the maximum amplitude of v, namely V_m.



- In summary, we can completely describe a specific sinusoidal signal if we know its frequency, phase angle and amplitude (either the maximum or the rms value).

The Sinusoidal Source

1

- Example-1: A sinusoidal current has a maximum amplitude of 20 A. The current passes through one complete cycle in 1 ms. The magnitude of the current at zero time is 10 A.

  a) What is the frequency of the current in hertz?

  b) What is the frequency in radians per second?

  c) Write the expression for $i(t)$ using the cosine functions. Express φ in degrees.

  d) What is the rms value of the current?

Solution: a) T is given in the question as 1 ms. $f = 1/T = 1/0.001 = 1000\ Hz = 1\ kHz$

b) $\omega = 2\pi f = 2000\,\pi = 6284$ rad/s

c) The magnitude of the current at zero time is 0.5 of the maximum value. $cos^{-1}(0.5) = 60°$

$i(t) = I_m \cos(\omega t + \varphi) = 20\cos(6284t + 60°)$

d) The rms value of a sinusoidal current is $I_m/\sqrt{2} = 20/\sqrt{2} = 14.14\ A$

The Sinusoidal Source

1

- Example-2: A sinusoidal voltage is given by the expression $v(t) = 300\cos(120\pi t + 30°)$

  a) What is the period of the voltage in milliseconds?

  b) What is the frequency in hertz?

  c) What is the magnitude of v at $t = 2.778\ ms$?

  d) What is the rms value of v?

Solution: a) From the given expression for v, $\omega = 120\pi$ rad/s

$\omega = 2\pi/T \qquad T = 2\pi/\omega = \frac{1}{60}\ s = 16.667$ ms

b) The frequency is $1/T = 60\ Hz$

c) Considering $\omega = 2\pi/16.667$, at $t = 2.778\ ms$

$\omega t = \frac{2\times\pi\times2.778\times10^{-3}}{16.667\times10^{-3}} = 1.047\ rad = 60° \qquad v(2.778\ ms) = 300\cos(60° + 30°) = 0\ V$

d) $V_{rms} = 300/\sqrt{2} = 212.13$ V

End of the Lesson (25.12.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist